

Mostofa Aadeeb Chowdhury

UNDERGRADUATE RESEARCHER, MECHANICAL ENGINEERING · KHULNA UNIVERSITY OF ENGINEERING AND TECHNOLOGY

mostofaaadeebchowdhury.me.kuet@gmail.com | [Google Scholar](#) | [LinkedIn](#)

Summary

Final-year Mechanical Engineering student at KUET (Graduation: August 2026) with a strong research portfolio spanning ML-guided design, composite materials, auxetic metamaterials and additive manufacturing. Published two Q1 journal papers (Composites Part B, Materials & Design) with five manuscripts under review or revisions submitted across Elsevier, Springer Nature and EAAI. Currently mentoring a 15-student undergraduate research team in literature review, research framing, and academic writing while coordinating between the team and an overseas faculty supervisor. Hands-on experience with 3D printing, CNC, UTM, and SEM, alongside simulation (ANSYS, Abaqus) and ML tools (Python, XGBoost, SHAP). Seeking a Lecturer position in Mechanical Engineering or a relevant field to contribute to undergraduate teaching and sustain an active research agenda.

Education

Khulna University of Engineering and Technology (KUET)

Khulna, Bangladesh

B.Sc. IN MECHANICAL ENGINEERING

2022 - 2026

- CGPA: 3.64/4.00 (through 7th semester) · Class Rank: 8/120
- Expected graduation: August 2026
- Thesis Title: *Auxetic-Clamping Suppression of Peel-Stress Failure in FDM-PETG Single-Lap Adhesive Joints: A Closed-Loop, Machine-Learning-Guided Experimental Design Study*

Publications

UNDER REVIEW/REVISION SUBMITTED

- **M. A. Chowdhury**, S. Chowdhury, M. A. Islam*, and M. G. Kader, "Auxetic-Clamping Suppression of Peel-Stress Failure in FDM-PETG Single-Lap Adhesive Joints: A Closed-Loop, Machine-Learning-Guided Design Study," submitted to *Engineering Applications of Artificial Intelligence*, 2026.
- **M. A. Chowdhury**, S. Chowdhury, M. M. Alam, and M. A. Al Bari*, "Public Acceptance of Nuclear Energy: A Systematic Review and Integrative Framework," submitted to *Progress in Nuclear Energy*, 2026.
- **M. A. Chowdhury**, R. Ahammad, H. R. Paul, S. Roy, N. U. H. Durjoy, A. D. Sarker, and M. M. Rahman*, "Additive Manufacturing and 3D Printing of Sandwich Structures," Ch. 11 in *Sandwich Structured Composites: Materials, Design and Applications*, Elsevier, 2026.
- **M. A. Chowdhury**, R. Ahammad, H. R. Paul, S. Roy, N. U. H. Durjoy, A. D. Sarker, and M. M. Rahman*, "Sandwich Composites in Automotive, Rail, Aerospace, UAV, and Marine Applications," Ch. 29 in *Sandwich Structured Composites: Materials, Design and Applications*, Elsevier, 2026.
- **M. A. Chowdhury**, S. Chowdhury, M. Rahman, R. Ahammad, Md. M. Islam, Md. N. Uddin, and M. M. Rahman*, "Comparative Analysis of CFRP vs. Metal Alloys in Structural Applications," Ch. 42 in *Handbook of Carbon Fiber Reinforced Polymer Composites*, Springer Nature, 2026.

JOURNAL

- A. Baral, **M. A. Chowdhury**, F. A. Rafi, Md. A. Al Khosru, S. Hossain, S. Chowdhury, and M. A. Islam*, "A diverse and explainable machine learning framework for multi-property prediction in nanoparticle-enhanced carbon fiber/epoxy composites," *Composites Part B: Engineering*, vol. 312, Art. 113310, 2026.
- A. Baral, **M. A. Chowdhury**, Md. J. Hasan, A. A. Nir, M. B. Arnab, and M. A. Islam*, "Prediction of mechanical properties of carbon fiber/epoxy composite modified by nanoparticles using multiple explainable machine learning algorithms," *Materials & Design*, vol. 257, Art. 114537, Aug. 2025.

• **CONFERENCE**

- M. M. Alam, S. Chowdhury, **M. A. Chowdhury**, M. M. Hasan, M. A. H. Shovon, and M. A. Al Bari*, "Social Acceptance of Nuclear Power: Insight from Qualitative Interviews in Bangladesh," in *Proc. SCOPE, Saudi Int. Conf. on Nuclear Power Engineering*, KFUPM, Dhahran, Nov. 2025, Paper 25199.
- J. Mondal, M. M. Alam, M. M. Hasan, **M. A. Chowdhury**, S. Chowdhury, M. A. H. Shovon, and M. A. Al Bari*, "Investigation of VVER-1200 under steam line break containment and partial turbine failure," in *Proc. SCOPE, Saudi Int. Conf. on Nuclear Power Engineering*, KFUPM, Dhahran, Nov. 2025, Paper 25219.

Research Experience

Undergraduate Thesis – Department of Mechanical Engineering, KUET

Khulna, Bangladesh

Auxetic-Clamping Suppression of Peel-Stress Failure in FDM-PETG Single-Lap Adhesive Joints: A Closed-Loop, Machine-Learning-Guided Experimental Design Study

SUPERVISOR: MD. GOLAM KADER / Co-SUPERVISOR: DR. MD. ASHRAFUL ISLAM

Sept. 2025 – Aug. 2026

- 3D-printed FDM-PETG auxetic-clamping inserts (negative Poisson's ratio geometries) to suppress peel-stress concentration at single-lap adhesive joint overlap edges.
- Ran a large-scale experimental campaign across design variations; measured failure load and mode; used RF, XGBoost, and Bayesian optimization to improve geometry iteratively based on experimental feedback.
- Identified key geometric parameters governing peel-stress suppression via SHAP explainability; manuscript under review at *Engineering Applications of Artificial Intelligence* (EAAI-26-21519).

LPBF Residual Stress & Distortion Prediction (ML Framework)

Khulna, Bangladesh

SUPERVISOR: DR. MD MAHBUBUR RAHMAN

July 2026 – Present

- Building a parametric ANSYS FEA study of Ti-6Al-4V LPBF across laser power, scan speed, layer thickness, hatch spacing, and part geometry to generate structured ML training data.
- Training and benchmarking RF, XGBoost, Gradient Boosting, and ANN models for simultaneous residual stress and distortion prediction; applying SHAP analysis for process parameter sensitivity ranking.

Mechanical Performance of GFRP Epoxy Laminates

Khulna, Bangladesh

SUPERVISOR: DR. MD. ARIFUZZAMAN

Oct. 2024 – Present

- Fabricated laminates with nine nanofillers (Al_2O_3 , CuO, TiO_2 , SiO_2 , nanoclay, graphene, pearlite, fly ash, MgO) across five loadings (0.1–0.9 wt%); measured tensile and flexural properties with SEM, XRD, and fracture analysis.
- Validated the whole experiment for the second time to ensure reproducibility.

Mechanical Properties Prediction of CFRP Composites

Khulna, Bangladesh

SUPERVISOR: DR. MD ASHRAFUL ISLAM

Sept. 2024 – Dec. 2025

- Trained RF, XGBoost, Gradient Boost, LightGBM, KNN, SVR, and Decision Tree models on nanoparticle-enhanced CFRP datasets; used SHAP to rank feature importance across tensile and flexural outputs.
- Identified fiber-layer count as the dominant tensile predictor and curing duration as the key flexural driver; proposed data-guided manufacturing settings for better laminate performance.

Nuclear Energy Research: Social Acceptance & Safety Analysis

Khulna, Bangladesh

SUPERVISOR: DR. MD ABDULLAH AL BARI

July 2024 – January 2026

- Synthesized scholarship on risk perception, institutional trust, social costs, and media dynamics around nuclear power; drew cross-country contrasts (Japan, China, Germany, France, developing nations).
- Found institutional trust to be a stronger predictor of acceptance than technical risk metrics; proposed communication pathways.
- Simulated 25–50% steam-line breaks plus 50% turbine trip with PCTran (VVER-1200); observed scrams at ~3 s, ~21 s, ~91–95 s; system restabilized ~1000 s – accepted, SCOPE 2025 (KFUPM).

Teaching & Mentoring Experience

2025–Present	Research Coordinator & Student Mentor <i>Composites, Vitrimers & Photocatalytic Materials research group of Dr. Abdullah Al Bari (KFUPM) – KUET Mechanical Engineering batches 2k21 & 2k22</i>	<i>Remote (Khulna, Bangladesh)</i>
	· Serve as on-the-ground research coordinator for the student team while the supervising faculty is based in Saudi Arabia. · Delivered training sessions on research fundamentals for 15 students: framing a research question, core terminology, reference management with Zotero, the structure of a research paper, and how to read literature to identify research gaps. · Acting as the primary point of contact between the students and the supervisor: resolve routine questions directly, escalate substantive issues for the supervisor's decision, schedule and run progress-review meetings, and relay student updates back for further direction.	

Professional Experience

2025	Intern, RAK Ceramics Completed a professional internship in industrial/manufacturing operations; awarded a certificate of completion.	<i>Gazipur, Bangladesh</i>
2024–2026	Student Operator, Fabrication Lab (FabLab), KUET Maker-space operations: assisted with 3D printers, laser cutter, CNC mill, vinyl cutter, and PCB mill; delivered training workshops and enforced lab safety.	<i>Khulna, Bangladesh</i>

Projects

Design, Assembly & Performance Test of Quadcopter for Surveillance & Aid Delivery (Mechanical Engineering Project, ME 3200)		2025
Designed, assembled, and flight-tested a quadcopter concept for surveillance and aid-delivery missions as a credited mechanical engineering project.		
Design of a Winch System for Construction Cranes (Machine Design II, ME 3217)		2024
Designed a winch system for construction cranes, sizing drum, gearing, and cable for rated load and duty cycle as a Machine Design II sessional.		
Design of a Hydrostatic Bursting Pressure Test Setup (Machine Design I, ME 3117)		2024
Designed a hydrostatic bursting-pressure test setup for pressure-vessel qualification as a Machine Design I sessional.		
CFD Simulation through a Centrifugal Pump (SolidWorks, SimScale)		2023
Used SolidWorks for CAD and SimScale to analyze fluid flow; investigated laminar vs. turbulent regimes.		
A Line Follower Robot for the Intra-KUET LFR Competition (Arduino)		2022
Used Arduino to follow straight/curved tracks reliably in the arena and won the competition.		
Computational Fluid Mechanics – Airflow around a Spoiler (SimScale)		2022
Ran an incompressible flow simulation around a spoiler using SimScale.		
Finite Element Analysis: Convergence and Mesh Independence (SimScale)		2022
Performed a mesh convergence study using SimScale.		
FEM – Linear, Nonlinear Analysis and Post-Processing (SimScale)		2022
Ran linear and nonlinear static analyses and post-processing in SimScale.		

Grants, Awards & Honors

2024–2025	University Dean's List Award for 3rd Year – KUET Authority
2025, 2026	National Undergraduate Math Olympiad, Bangladesh – 3rd Place (2026), 5th Place (2025) – Bangladesh Math Society
2022, 2023	International Youth Math Challenge – Silver Honor (93rd percentile) – IYMC
2013–2020	Bangladesh Mathematical Olympiad (BdMO) – National & Regional Winner and IMO Camper – BdMO
2018	Bangladesh Physics Olympiad (BdPhO) – Divisional Winner – BdPhO
2015–2017	National High School Programming Contest, Bangladesh (NHSPC) – Divisional Champion – NHSPC
2015–2023	Education Board Talentpool Scholarship (HSC, SSC & JSC) – Education Ministry of Bangladesh

Affiliations & Volunteering

2025–2026	Human Resource Officer – IMechE Student Branch, KUET Organized the SOFE (Society of Future Engineers) article-writing competition.
2024–2025	Executive – IMechE Student Branch, KUET
2025–2026	General Secretary – KUET Math Club Led and organized Multiverse of Mathness 2.0 and the Intra-KUET Integration Bee & Puzzle Competition.
2024–2025	Treasurer – KUET Math Club
2023–2024	Senior Executive – KUET Math Club
2022–2023	Junior Executive – KUET Math Club
2022–2024	Academic Team Member – Bangladesh Physics Olympiad Led the academic team in BdPhO Khulna Region (2023); volunteered for question setting, moderation, and script checking since 2022.
2022–2025	Academic Team Member – Bangladesh Mathematical Olympiad Volunteered for question setting, moderation, and script checking since 2022.
2022	Project Lead – Team Five Wheel Led a 5-member team to build a Line Following Robot (competition champion).
2016–2018	Co-Founder – Sigma Math Club, Jashore Trained junior & elementary students for math Olympiads; 17 students won competitions, 2016–2018.

Test Scores

IELTS: 7.0 (L-8, R-7, W-7, S-6.5)

Sept. 2021

Technical Skills

- **Programming & ML:** Python (PyTorch, Pandas, NumPy, Matplotlib, Scikit-learn), C, ML applications
- **Simulation & CAD:** Abaqus, Ansys, SolidWorks
- **Hardware & Fabrication:** 3D Printing, Arduino, UTM, SEM, CNC & Laser Cutting

References

Dr. Md. Ashraful Islam

PROFESSOR, MECHANICAL ENGINEERING

Khulna University of Engineering and Technology

md.islam@me.kuet.ac.bd

[Webpage](#) | [Scholar](#)

Dr. Md. Abdullah Al Bari

ASST. PROFESSOR, MECHANICAL ENGINEERING

King Fahd University of Petroleum & Minerals

mdabdullahal.bari@kfupm.edu.sa

[Webpage](#) | [Scholar](#)